ASSIGNMENT 1

Your team, your agent, your problem

FlowAssist — Agent-assist copilot for B2B payments

Data & AI Product: From Diagnosis to Delivery · Albert School / Mines Paris-PSL
Lenny Santus · Anthony Rascle · Antoine Rabeau · Benjamin Caujolle · Colin Sibieude
PDF edition · 6 November 2026 · Coursework in English, following the supplied briefs.

Team GitHub: @lsantus-mines · @antotototototo18 · Antoine @antorabo75 · @BenjaminAHS · Colin @csib-wq

Team and agent

Member	GitHub
Lenny Santus	@lsantus-mines
Anthony Rascle	@antotototototo18
Antoine Rabeau	@antorabo75
Benjamin Caujolle	@BenjaminAHS
Colin Sibieude	@csib-wq

Company / proposal: PayFlow B2B, European B2B payments fintech; P2 adapted as FlowAssist. Course scenario: 800 merchants, 12,000 tickets/month, 40% of handling time spent investigating. Other operating figures below are planning assumptions, not verified company observations.

Job: When a merchant opens a payment support ticket, FlowAssist retrieves authorised transaction evidence and produces a cited draft in the Zendesk sidebar for a human specialist to review, edit and send.

Authority: L1 entity suggestions → L3 tenant-scoped read-only retrieval, human confirms match → L2 drafting/checks → person-only sending; L4 bounded audit metadata. The backend establishes identity; model confidence cannot authorise access.

Never: send to a merchant, move/refund/release funds, write the ledger, or disclose a transaction without verified tenant and transaction identity. Ambiguity, stale data or unsupported intent suppress drafting and return to a person.

Problem in one sentence

Tier-1 payment specialists spend an assumed 7.2 of 18 minutes per ticket investigating disconnected records, equivalent to €414,720/year in investigation labour; today they search manually and draft from memory. Success targets a 30% AHT reduction to 12.6 minutes, QA ≥92% and zero critical data exposures.

Assumption register

Figure	Source type / validation
800 merchants; 12,000 tickets/month; 40% investigation	P2 educational scenario, not actual CRM/ticket logs
144,000 tickets/year	Derived: 12,000 × 12
18-minute AHT; 4.5-hour first response; 18 agents	Guesses; validate with Ops logs and paired timing
7.2 investigation minutes	Derived: 18 × 40%
€24/hour; €10/ticket	Planning analogies; wage/cost allocation unverified; do not treat them as identical bases
€414,720 investigation value	144,000 × 7.2 / 60 × €24; capacity estimate
30% saving, QA ≥92%, zero exposure	Product decisions/targets, not measured results

Persona: Karim, Tier-1 specialist, needs to explain a payment without confusing merchant identity, clearing status or bank credit. Ops owns quality, CFO owns investment, DPO/CCO owns data/compliance review. The task is frequent, bounded and reversible before sending.

02 · Diagnosis and AI choice

Journey and the two pain steps

Human activity today	Minutes (guess)	Planned assistance
Read ticket	1.5	L1 entity suggestions
Search ledgers / clearing — pain	5.7	L3 scoped candidates, person confirms
Draft and attach sources — pain	4.8	L2 cited draft
Check policy / status	2.5	Rules block; L2 review
Review, edit, send	2.0	Person only
Categorise and log	1.5	L1 route; L4 audit metadata
Total	18.0	Assumed baseline; not measured

Problem tree and five Whys

Symptom: merchants wait for payment explanations; specialists switch repeatedly between tools. **Problem:** evidence gathering and reply composition are fragmented. 1. Why slow replies? Specialists investigate before drafting. 2. Why lengthy investigation? Ledger, clearing, dispute and compliance records sit in separate interfaces. 3. Why separate interfaces? Independent services lack a unified support view. 4. Why not one exact lookup? Incoming language often has partial references, dates and gross/net amounts. 5. Why retain a person? Ambiguous identity, status and intent create costly disclosure and financial mistakes.

Root cause and fix: fragmented evidence and unstructured requests. First add deterministic read-only joins, source visibility and clear escalation. Use extraction and grounded drafting only where rules/templates are insufficient; improve the workflow and documentation alongside the AI.

Is AI the right tool?

Required question	Decision
Does a rule solve 80%?	Not established. Benchmark exact references, rules and templates first; the earlier 15% claim is unverified.
Prediction/generation or organisation?	Entity extraction and cited wording may need a pretrained model. Permission, money and escalation policy are deterministic/ organisational.
Do we have the data?	Only synthetic fixtures. Future read-only ledger, clearing and policy access needs owner and DPO approval; no 144k real-ticket corpus supplied.
What does an error cost?	S2 judgement: €15 tracing, €90 disputes, €150 compliance; catastrophic disclosure tail remains unpriced. PayFlow owns organisational liability.

Chosen level: bounded GenAI copilot. Rules remain mandatory; classical classification alone cannot compose a cited multi-source explanation. A standalone pretrained model lacks current ledger facts. Autonomous customer action is unjustified; compare hybrid drafting with rules/templates in S4.

Four selection questions

Frequency: course scenario 12,000/month. **Bounded:** read-only account, ledger, clearing and compliance tools; draft, context-only or specialist route. **Measurable:** timing, factual QA, grounding and critical failures per ticket. **Recoverable:** suppress or correct before sending; human review reduces risk but does not guarantee detection.

03 · Hypotheses and baseline

Working backwards — fictional launch release

PARIS — 18 November 2026 (fictional launch exercise). PayFlow announces FlowAssist for payment support specialists: a verified payment timeline and cited reply inside Zendesk, targeting an AHT reduction from 18 to 12.6 minutes while the specialist retains control. “I can see the evidence before I answer,” says a fictional merchant representative. This quotation and the claimed future improvement are invented solely for the required press-release format, not interview evidence or achieved performance.

Three hypotheses that must be true

Hypothesis (guess)	Test and session
≥75% of tracing tickets contain enough evidence for a unique match with ≥99% matching recall	S4: 200 independently labelled cases; compare extraction + deterministic lookup with rules alone; count ambiguous and missed matches.
Grounding checker flags unsafe drafts with ≥98% precision	S5: independently grade safe/unsafe draft pairs, including adversarial mismatches; also report unsafe recall and false-safe count.
≥30% overall AHT saving without QA falling below 92% or 7-day reopen exceeding 5%	S6: planned agent-level randomised trial, 9 assisted / 9 control specialists, four weeks; log active review and fallback time. Staffing, thresholds and duration are planning decisions.

Economic test connecting S2/S3: ≥41.2% paired active-time saving on pre-labelled eligible tracing cases; this is different from 30% across the full support mix. A tracing-only pass does not validate dispute/compliance value or production readiness.

Do nothing over 12 months

Assumption	Consequence / source type
4% natural efficiency gain through macros and documentation	€57,600/year = 144k × €10 × 4%; planning analogy, not verified trend
Merchant/ticket growth +25%	1,000 merchants and 15,000 tickets/month; scenario guess
Four additional hires at €55k loaded cost	€220k/year staffing scenario; do not add again to S2 benefits
€35k baseline SLA penalties; +40% absent intervention	€49k/year scenario; contracts and attribution unverified

The baseline is not zero. S2 compares FlowAssist with this €57,600 efficiency value and the named €125,000/year Merchant Self-Serve Dispute & Reconciliation Portal. Avoid counting growth, hiring avoidance and time savings as independent benefits without evidence.

P2 adaptation and scope

FlowAssist keeps P2's support investigation job. Initial MVP handles payment tracing; disputes, fraud and compliance route to specialists. No custom-idea pitch is required when using P2; the optional original pitch remains in the source archive.

04 · Governance and submission

Six provisional governance answers

Question	Specific answer and check owner
EU AI Act tier	Internal tracing support; no credit/eligibility decision. Classification and applicable transparency duties require Legal review of actual intended use. Label drafts as AI-assisted.
Whose data / legal basis?	Merchant contacts and transaction data, currently synthetic only. DPO must assess purpose, data-subject relationship, contractual necessity or legitimate interests as applicable; no blanket B2B contract approval.
Accountability / human	Head of Payment Ops owns answer quality; human specialist reviews/sends; Engineering enforces access and gates. Organisation remains accountable.
What must it never do?	No customer sending, ledger/fund writes or unverified tenant/ transaction disclosures. Model score never overrides a block.
Explainable to whom?	Specialist: facts, IDs, source timestamps and block reasons. Merchant: clear verified status. Authorised auditor: minimised versioned approval/gate metadata.
People it cannot serve?	Ambiguous/stale/missing evidence → context-only/manual handling. Dispute, KYC/fraud and unsupported language → named specialist, preserving authorised context.

Cost-of-error ownership

Routine rework is priced in S2, but wrong tracing matches can also expose financial data. Dispute misinformation and compliance disclosures create legal/reputational harm beyond operational expected losses. **CCO/DPO** assess data/legal incidents; **Ops** owns quality and specialist routing; **Engineering** owns deterministic authorization and draft shutdown. A second LLM is not a security guarantee.

Current control requirements

Tenant authorization must precede retrieval. Use parameterised read-only queries, exact transaction checks, redaction, source timestamps and human approval. Wrong-transaction disclosure, tenant exposure or attempted financial/customer action immediately disables drafting; Ops stops, Engineering enforces, DPO/CCO assesses the incident. Live access, provider terms, hosting, no-training/retention and legal classification remain to approve. No compliance certification is claimed.

Submission and evidence boundary

All numerical operating inputs above are course facts, derived values, decisions, analogies or guesses as marked; no actual PayFlow exports, time study, interviews or signed production terms are available. These statements supersede unsupported historical “measured” claims in the archived source files.

The companion archive preserves README.md, 01_problem_statement.md, governance.md and the optional 00_pitch.md. The assignment requires at least two members to commit with their own accounts; PDF generation cannot supply or certify those contributions. Team alignment with Agentic Workflows must be confirmed by the team; the README currently records “no” for all five members.